

KAUSHAL G BHAT

Full-Stack Software Engineer | Architecture & AI Specialist

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PROFILE

Full-stack software engineer specializing in TypeScript, Python, and Dart/Flutter ecosystems with deep expertise in architecting multi-tenant SaaS platforms, AI integrations, and domain-driven systems. Proven track record of delivering end-to-end products across healthcare, e-commerce, logistics, fintech, and mobile domains. Committed to clean architecture, comprehensive testing, and production-ready code that scales.

TECHNICAL SKILLS

- Languages:** TypeScript, Python, Dart, SQL
- Frontend:** React 19, Next.js 16, Flutter, Tailwind CSS v4, Framer Motion, GSAP, Radix UI, Shadcn UI
- State & Data:** Zustand, TanStack Query, React Hook Form, Zod, Riverpod
- Backend:** FastAPI, Flask, Next.js Server Actions, Node.js
- Databases & ORM:** PostgreSQL, SQLite, Supabase, Prisma, Drizzle ORM, SQLAlchemy, Drift
- AI/ML:** OpenRouter, Multi-provider AI abstraction, Optical Character Recognition (OCR), Prompt engineering
- Auth & Security:** NextAuth, Better Auth, JWT, Supabase Auth, Role-Based Access Control (RBAC), Row-Level Security (RLS), HMAC verification
- Payments:** Razorpay (webhooks, subscriptions)
- Testing:** Vitest, Playwright, pytest, MSW
- DevOps & Tools:** Vercel, GitHub Actions, Alembic, esbuild, npm workspaces, Husky, Commitlint
- Architecture:** Clean Architecture, Domain-Driven Design (DDD), Repository Pattern, Event-Driven, Multi-tenant SaaS, Command Query Responsibility Segregation (CQRS)

PROJECTS

Legacy Modernization Platform — COBOL to NestJS Migration Engine | *TypeScript, React 19, Express v5, Neo4j, Zustand, React Query, Vitest, Zod, Web Workers, Docker*

Universal mainframe modernization platform that discovers, analyzes, transforms, validates, and certifies migration of COBOL/CICS/VSAM applications to NestJS/TypeScript microservices

- Architected 39-stage directed acyclic graph (DAG) pipeline with 10+ TypeScript engine packages — COBOL parser, Intermediate Representation (IR) engine, pattern detection (12 types), NestJS code generator (21 templates), and multi-dimensional equivalence verification
- Built knowledge graph system (Neo4j, 291 nodes, 661 relationships) with 6 intelligence analyzers, config-driven multi-system universalization (scored 90/100 across 4 unrelated legacy systems with zero code modifications), and production certification framework (V1/V2/Master)
- Developed dual React 19 frontends — a 10-page read-only Explorer dashboard backed by Express v5 REST API (23 endpoints), and a 19-page Studio with Zustand (11 slices), browser-based Web Worker pipeline execution, and full artifact registry
- Integrated context-aware AI agents that dynamically update the platform as new COBOL applications are ingested — enabled by clean architecture and modular coding patterns allowing on-the-fly system evolution without downtime

Creatos — Website Operating System | *Next.js 14, Prisma, Supabase, Razorpay, NextAuth, Framer Motion, TypeScript*

Multi-tenant SaaS platform for creators/agencies to generate and publish websites with AI

- Architected registry-driven domain system with 10+ bounded contexts and snapshot-based publishing pipeline with Incremental Static Regeneration (ISR) cache invalidation
- Built marketplace for themes/templates (30 themes, 8 categories) with capability-driven feature gating
- Implemented Razorpay subscription billing, white-label agency support, and multi-role access (Creator, Agency, Enterprise, Super Admin)

JOBinder — AI-Powered Job Discovery | *Next.js 16, React 19, Supabase, Zustand, TanStack Query, Zod, Vitest, Playwright*

AI-powered job discovery platform with Tinder-style swipe interface

- Designed clean architecture with domain-driven modules, repository pattern (Supabase adapter), and use-case-driven business logic
- Built multi-provider AI abstraction layer (adapter pattern) for resume parsing, job scoring, and ranking

- Implemented swipe-based discovery feed with optimistic UI, referral system, and real-time WebSocket updates

Nevase Dental — Dental Practice Management | *FastAPI, SQLAlchemy, PostgreSQL, Alembic, React, TypeScript, Vite, JWT*

Full-stack dental clinic management system for patients, appointments, billing, and clinical documentation

- Developed full appointment lifecycle engine (scheduling → check-in → vitals → prescription → completion) with electronic prescription generation
- Implemented SOAP notes documentation, vital signs tracking, and structured audit logging for clinical compliance
- Built billing/invoicing system, inventory management, and role-based access control (admin, doctor)

ShroomCulture — Premium E-commerce | *Next.js 16, Supabase, Zustand, React Query, Framer Motion, GSAP, Lenis, Tailwind CSS v4*

Premium organic mushroom e-commerce with sophisticated animations and brand-first design system

- Architected Domain-Driven Design with repository pattern (factory-based Mock/Supabase switching), specification pattern, and value objects
- Implemented strict layering where database rows never reach UI — mapper layer converts to domain models exclusively
- Built custom cursor system, scroll progress tracking, GSAP pinned animations, and View Transitions API page transitions

ScreenFix AI — Android AI Assistant | *Flutter 3.29, Dart, Riverpod, GetIt, GoRouter, Dio, Drift, OpenRouter API*

Android AI assistant that captures screens, extracts text via OCR, and provides AI-powered recommendations through a floating overlay

- Architected clean architecture with feature-first vertical slices, dependency injection (GetIt), and cross-feature integration layer
- Built end-to-end AI pipeline: screen capture → OCR extraction → pattern pre-filter → structured prompt builder → OpenRouter inference
- Implemented floating overlay system, anonymous device identity, feature flags, and telemetry event tracking

WaariPG — PG/Hostel Management | *Next.js 16, React 19, Supabase, Shadcn UI, React Hook Form, Zod, Framer Motion*

PG and hostel management platform with resident tracking, room allocation, gate logging, and payments

- Built feature-based modular architecture with RBAC, repository pattern, and service layer
- Implemented resident management, room allocation, gate log tracking, payment processing, and analytics dashboard
- Added command palette (cmdk), onboarding flow, product tour, notifications, and help center for enhanced UX

Milk Delivery — Route Management | *Next.js 16, Drizzle ORM, NeonDB, Better Auth, DnD Kit, TanStack Table*

Milk delivery route management platform for daily operations, drivers, and customer delivery scheduling

- Designed delivery plan engine supporting 4 types (daily, alternate days, custom weekdays, paused/vacation) with automatic route exclusion
- Built drag-and-drop route stop ordering, per-driver/per-route delivery dashboards, and tenant-scoped CSV exports
- Implemented multi-tenant architecture with Better Auth, NeonDB serverless PostgreSQL, and Google Maps integration

Payment Automation — Razorpay Webhooks | *Python, Flask, Razorpay API, Google Sheets API, Streamlit, pytest*

Webhook automation capturing Razorpay payment events with signature verification and Google Sheets logging

- Built Hash-based Message Authentication Code (HMAC)-verified webhook handler for payment.captured events with normalized duplicate detection
- Implemented Streamlit analytics dashboard with auto-refresh, batch/mode filters, CSV export, and revenue insights
- Designed for production with gunicorn deployment, IST timezone handling, and retry-safe idempotent processing

Other Projects

- 3D Printables — Next.js e-commerce with HEIC→JPEG image optimization, NextAuth v5, and SEO (sitemap/robots.txt)
- Help Indian Football — Civic advocacy petition platform with Supabase, Resend email notifications, and role-based portals
- MicroNest — Project management tool with browser extension (esbuild bundled), virtual scrolling, and Excel export
- MicroNest Services — Design system monorepo analyzing SaaS marketing patterns (Linear, Intercom, Mintlify)
- Medical App — Clinic operations system with front desk dashboard, nurse workflows, and queue management

WORK EXPERIENCE

Operations Head — Money Craft Trader*Jul 2025 – Present*

- Manage end-to-end operations for stock-market education programs, resolving 150+ customer queries daily across WhatsApp and email
- Implemented WATI WhatsApp automations, cutting manual workload and response time, and improving live-session sit rate by 5–8%
- Coordinate with marketing and trainers to run smooth batch execution; own the company's LMS and product operations

Operations Executive — Goenka Kachave LLP*Dec 2023 – Jun 2025*

- Provided customer support via WhatsApp and email for educational products, resolving payment, onboarding, and LMS-access issues
- Maintained SOP documentation for recurring support issues and ensured SLA adherence and customer satisfaction

Digital Marketing & Graphic Designer — Ambroasian Research and Development *Nov 2021 – Oct 2023*

- Designed marketing creatives for social media campaigns and supported lead engagement and execution
- Collaborated with operations teams to align communication across campaigns

EDUCATION

Bachelor of Computer Science — Sinhgad College of Science, Arts and Commerce (2015 – 2019)

Advanced Program in Interactive Design & Games — Maya Academy of Advanced Cinematics (2020 – 2023)